

Learning Management System (LMS) Documentation

1. Project Overview

Project Name: Learning Management System (LMS)

Technology Stack:

- Backend: Node.js (Express.js)
- Frontend: React.js (Admin & Client Side)
- Database: MongoDB / MySQL
- Authentication: JWT & Role-based Access Control
- Hosting: AWS / Vercel / Netlify

2. Modules Overview

Courses:

- View All Courses
- Live Classes
- Text Courses
- In-App Live History
- New Course
- Course Notes

Course Bundles:

- List Bundles
- Create New Bundle

Upcoming Courses:

- List View

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- Add New Upcoming Course

Course Notices:

- Notice List
- Add New Notice

Certificates:

- Quiz Certificates
- Completion Certificates
- Certificate Templates
- Create New Template
- Settings

Quizzes:

- Create Quizzes
- Evaluate & Grade

3. Security & Anti-Cheating Features

Authentication & Authorization:

- JWT-based login system
- Role-based permissions
- Two-Factor Authentication (Optional)

Data Security:

- HTTPS

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- Rate limiting & brute force protection
- Input validation and sanitation
- Secure file uploads

Screen Security (Web Limitations Apply):

- Prevent screen recording (Desktop App/DRM)
- Prevent screenshots (watermarking & overlays)
- Session tracking (tab switch monitoring, disable right-click/shortcuts)

4. Admin Panel Features

- Dashboard Overview
- Course and Bundle Management
- Student & Instructor Management
- Certificate Settings
- Quizzes and Grading Management
- Course Notice Handling

5. Frontend Features

- Browse/Enroll in Courses
- View Live and Text Courses
- Access Course Notes
- View Certificates
- Join Live Classes
- Receive Notices and Announcements

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6. Future Enhancements

- Video DRM & Tokenized Access
- AI-based Proctoring
- Mobile App Support
- Notifications (Email & In-App)

7. UI Screenshots

Attach screenshots for: Navigation Sidebar, Course Management, Certificates Panel, Bundles and Notices.

8. Screen Security Recommendations

- Use desktop app with Electron to restrict OS-level controls
- Embed content with DRM support (e.g., Widevine)
- Add visible watermarks and dynamic overlays